

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (CE/IT)

December 2011

IT303 Advanced Database Management System

Date: 15.12.2011, Thursday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) What are the options of view? Explain advantages of views. [04]

(b) What is distributed transaction and how does it differ from remote transaction? Explain with an example. [03]

Q - 2 (a) Explain how a TP monitor manages memory and processor resources more effectively than a typical operating system. [06]

(b) Give the differences between client server architecture and a centralized computing system. [05]

(c) What is pragma EXCEPTION_INIT? Explain the usage. [03]

OR

Q - 2 (a) Explain why ambiguity potentially exists with multiple inheritances. Illustrate your explanation with an example. [06]

(b) List background processes with respect of transaction server. Explain any two of them in detail. [05]

(c) What is a cursor for loop? Explain the usage. [03]

Q - 3 (a) Discuss two-phase commit protocol for ensuring atomicity of distributed database. [05]

(b) Let relation r contain exactly the following tuples [05]

(6, 7, 8)
(UNK, 7, 8)
(6, UNK, 8)
(UNK, 7, UNK)
(UNK, UNK, UNK)

Assume, the three attributes are A, B, C of type integer. If V is a range variable that ranges over r, state the truth values of the following expressions:

- i) EXISTS V(V.B > 7)
- ii) EXISTS V(V.B > 9 AND V.C > 7)
- iii) EXISTS V(MAYBE(V.C < 6))
- iv) FORALL V(V.B > 1 OR IS_UNK(V.C))

- (c) State true/false, justify your answer. A linear job helps to exhibit parallelism in query execution. [04]

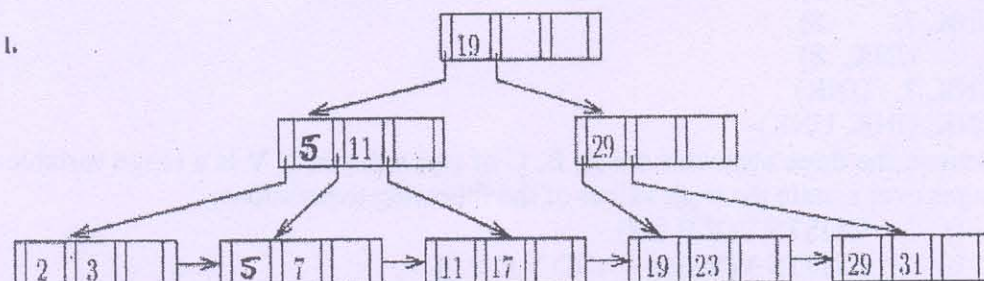
OR

- Q - 3 (a) What are the major DDAS components? Explain each in detail. [05]
(b) Explain Shared disk and Shared memory architecture. [05]
(c) Answer the followings: [04]
(1) To grant the SELECT privilege on the table EMP to all users in the database, which statement would you use?
(2) What do you mean by instance?

SECTION - II

- Q - 4 (a) Attempt the following: [04]
(1) What are the basic file organization methods? Explain any of them in brief.
(2) What are the three parts available in a trigger description? Explain with an example.
(b) Is it possible in general to have two primary indices on the same relation for different search keys? List as many reasons as possible. [03]

- Q - 5 (a) Answer the following: [08]
(1) In a range selection on a range-partitioned attribute, it is possible that only one disk may need to be accessed. Describe the benefits and drawbacks of this property.
(2) What is the purpose of compensating transactions? Present two examples of their use
(b) For the given B+-tree, show the form of the tree after each of the following series of operations: [06]
i) Insert 9.
ii) Insert 10.
iii) Insert 8.



OR

Q - 5 (a) Answer the following:

[08]

- (1) Compare RDBMS and ORDBMS. Describe an application scenario for which you would choose ORDBMS or RDBMS.
- (2) Why the bucket overflow occurs in hashing and how to handle bucket overflow problem?

- (b) Work through RSA public encrypting scheme with $p=3$, $q=11$, $e=9$, $d=9$ and plaintext="CHARUSAT". (Hint: assign 1, 2, ...26 to A, B, ...Z and perform the operation and write the ciphertext in alphabets. **[06]**

Q - 6 (a) Answer the following (Any **Three**)

[12]

- (1) State the difference between views and snapshot with example, in general what we are using in SQL.
- (2) Explain in brief oracle database objects.
- (3) What do you mean by authentication? Explain few authentication methods in database in brief.
- (4) Differentiate Homogenous and Heterogeneous Distributed Database System.
- (5) What are the methods for creating persistent objects?

- (b) Describe the drawbacks of pipelined parallelism.

[02]
